La Scuola d'Italia / Dr. Huson / IB Math 11
21 October 2026

Nickname:

2.3 Pre-Quiz: Sequences, Lines, Radicals and Exponents

1. Consider the geometric sequence

$$24,\ 12,\ 6,\ \ldots$$

   (a) Write down the common ratio r.

   (b) Find u_6.

   (c) Explain why the series $24 + 12 + 6 + \cdots$ has a sum to infinity.

   A different geometric series has first term 10 and sum to infinity 15.

   (d) Find the common ratio of this series.

2. Point P has coordinates $(-4,\ 1)$, and point Q has coordinates $(6,\ -3)$. Point M is the midpoint of $[PQ]$.

   (a) Find the coordinates of M.

   (b) Find the gradient of $[PQ]$.

   The line L is perpendicular to $[PQ]$ and passes through M.

   (c) Write down the gradient of L.

   (d) Find the equation of L.

   (e) The line L passes through the point $(k,\ 9)$. Find the value of k.

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3. Let $f(x) = 2x - 3$, for $x \in \mathbb{R}$.

   (a) Write down the y-intercept of the graph of f, and find the x-intercept.

   (b) The graph of a linear function g is parallel to the graph of f and passes through the origin. Find an expression for $g(x)$.

   (c) The graph of a linear function h is perpendicular to the graph of f and passes through the point $(2,\ 6)$. Find an expression for $h(x)$.

   (d) Find the coordinates of the point where the graphs of f and h intersect.

   (e) The graph of h meets the x-axis at the point R. Find the coordinates of R.

4. (a) The expression $4^x \times 2^{x+1}$ can be written in the form 2^{ax+b}. Find the value of a and the value of b.

   (b) Hence, solve the equation $4^x \times 2^{x+1} = 16$.

   (c) Solve the equation $9^x = 27$.

   (d) Show that $\left(\sqrt{12} + \sqrt{3}\right)^2 = 27$.

   (e) Write $\dfrac{6}{3 - \sqrt{3}}$ in the form $a + b\sqrt{3}$, where $a, b \in \mathbb{Z}$.